

Ngee Ann Secondary School
Secondary 4&5 Elementary Mathematics-O
2026 Prelim paper 1
Marking Scheme

Qn No.	Qn Part	Solutions	Marks (Remarks)	Total
1		$(3x-2)(8x^2-5x+1)$ $= 24x^3 - 15x^2 + 3x - 16x^2 + 10x - 2$ $= 24x^3 - 31x^2 + 13x - 2$	M1: Expansion A1	[2]
2	(a)	50 mins	B1	[1]
	(b)	She is incorrect as the number of patients who have waiting time less than 43 minutes is the same/equal as the number of patients who have waiting time more than 66 minutes.	B1	[1]
3		$81 = 3^4$ $108 = 2^2 \times 3^3$ LCM: $2^2 \times 3^4 = 324$	M1: either prime factorisation A1	[2]
4		$\frac{3(2k)}{4} = k - \frac{6(2k)-5}{5}$ $30k = 20k - 48k + 20$ $58k = 20$ $k = \frac{10}{29}$	M1: substitute x with 2k A1	[2]
5	(a)	2	B1	[1]
	(b)	9	B1	[1]
6	(a)	$15\pi - 24\pi b + 3\pi^2$ $= 3\pi(5 - 8b + \pi)$	B1	[1]
	(b)	$4x^2 + 9y^2 - 12xy - 64$ $= (2x - 3y)^2 - 64$ $= (2x - 3y - 8)(2x - 3y + 8)$	M1: $(2x - 3y)^2$ A1	[2]

7		$\frac{5x}{6} - \frac{3(x-1)}{4} + \frac{2x+7}{12}$ $= \frac{10x}{12} - \frac{9(x-1)}{12} + \frac{2x+7}{12}$ $= \frac{10x-9x+9+2x+7}{12}$ $= \frac{3x+16}{12}$	M1: common denominator A1	[2]
8	(a)	1.57 (3 sf)	B1	[1]
	(b)	11 and 12	B1, B1	[2]
9	(a)	547.39 billion $= 547.39 \times 10^9$ $= 5.4739 \times 10^{11}$	B1	[1]
	(b)	$\frac{5.4739 \times 10^{11}}{6.037 \times 10^6}$ $= \text{USD } 90\,672.52$	M1 A1	[2]
10		$3x + 5y = 4$ $7x - 4y = -22$ $12x + 20y = 16$ $35x - 20y = -110$ $47x = -94$ $x = -2$ $y = 2$	M1: Substitution/ Elimination A1 A1	[3]
11	(a)	Angle OAB is 90° because tangent is perpendicular to the radius.	B1	[1]
	(b)	Area of quadrilateral – Area of sector $= 6 \times 11 - \frac{126}{360} \times \pi(6)^2$ $= 26.4 \text{ cm}^2$	M1: either area A1	[2]
12	(a)	1 cm is to 60 km 24 cm is to 1440 km	B1	[1]
	(b)	1 cm^2 is to 3600 km^2 $1574722 \div 3600 = 437.4227778$ $= 437 \text{ cm}^2$	M1: area scale A1	[2]

13	(a)	$BD^2 = 20^2 = 400$ $BC^2 + CD^2 = 12^2 + 16^2$ $= 400$ Since $BD^2 = BC^2 + CD^2$, by Converse of Pythagoras' Theorem, it is a right angled triangle where BD is the hypotenuse. Hence BC is perpendicular to CD . OR $\cos BCD = \frac{12^2 + 16^2 - 20^2}{2(12)(16)}$ Angle $BCD = 90^\circ$ Hence BC is perpendicular to CD .	B1 B1	[1]
	(b)	$\cos x + \sin x$ $= -\cos CDB + \sin CDB$ $= -\frac{16}{20} + \frac{12}{20}$ $= -\frac{1}{5}$	M1: either value A1	[2]
14		$\frac{3x^2 + 7x - 40}{18x^3 - 128x}$ $= \frac{(3x-8)(x+5)}{2x(9x^2-64)}$ $= \frac{(3x-8)(x+5)}{2x(3x-8)(3x+8)}$ $= \frac{x+5}{2x(3x+8)} \text{ or } \frac{x+5}{6x^2+16x}$	M1: factorise numerator M1: factorise denominator A1	[3]
15		Angle $CDA = 132^\circ \div 2 = 66^\circ$ (angle at centre = 2 angles at circumference) Angle $CBA = 180^\circ - 66^\circ = 114^\circ$ (angles in opposite segment) Angle $CBP = 180^\circ - 114^\circ = 66^\circ$ (adjacent angles on a straight line)	B1 with reason B1 with reason B1: value of 66°	[3]

16		Size of interior angle : $(360^\circ - 60^\circ) \div 2$ $= 150^\circ$ Size of exterior angle: $180^\circ - 150^\circ$ $= 30^\circ$ Number of sides: $360^\circ \div 30^\circ = 12$ sides	M1: interior angle M1: exterior angle A1	[3]
17	(a)	The principal value increases every month, hence the appreciation value is not 1.2% of \$56700.	B1	[1]
	(b)	$58900 \left(1 + \frac{0.1}{100} \right)^{36}$ $= \$61057.93$	M1 A1	[2]
18		Let x be the cash price of the sofa bed. Total amount: $0.25x + 26 \times 88$ Interest: $(0.75x)(0.08)(26/12)$ $= 0.13x$ $1.13x = 0.25x + 2288$ $0.88x = 2288$ $x = \\$2600$	M1: Interest M1: correct equation A1	[3]
19	(a)	1 cupboard : 96 man- hours $96 \div (3 \times 4) = 8$ days $8 - 2 = 6$ days	M1: total number of days A1	[2]
	(b)	$y^3 = \frac{k}{x^2}$ where k is a constant $k \left(\frac{1}{4^2} - \frac{1}{8^2} \right) = 3$ $k \left(\frac{1}{16} - \frac{1}{64} \right) = 3$ $\frac{3}{64} k = 3$ $k = 64$ $x = 8$ or -8	M1 M1: Find k A1	[3]

20	(a)(i)	$\sqrt[3]{\frac{x^3 + 2y}{y}} = 2x$ $\frac{x^3 + 2y}{y} = (2x)^3$ $x^3 + 2y = 8x^3 y$ $x^3 = 8x^3 y - 2y$ $x^3 = y(8x^3 - 2)$ $y = \frac{x^3}{8x^3 - 2}$	M1: Cube on both sides M1: Terms with y on one side A1	[3]
	(a)(ii)	$y = \frac{(-2)^3}{8(-2)^3 - 2}$ $y = \frac{4}{33}$	B1	[1]
	(c)	$p = \frac{3}{q} + 2$ $pq = 3 + 2q$ $q(p - 2) = 3$ $q = \frac{3}{p - 2}$ It is inaccurate as q is undefined only when $p = 2$.	B1: Make q the subject B1	[2]
21	(a)	$58 - 30$ $= 28$	M1: either value A1	[2]
	(b)	$\frac{7}{12} \times 120 = 70$ Score: 50	B1: 70 B1	[2]
22	(a)(i)	$-\frac{12}{13}$	B1	[1]

	(a)(ii)	$\frac{5}{12}$	B1	[1]
	(b)	$\frac{1}{2}(5)(b) = 50$ $b = 20$ $20 - 3 = 17$ $R(17, 0)$	M1 A1	[2]
23	(a)	Since $(x - h)^2$ is always positive or zero for all values of x , the maximum value of y occurs when $(x - h)^2 = 0$. Hence, the maximum value happens when $x - h = 0$ or $x = h$.	B1	[1]
	(b)	$41 - 12x - x^2 = 0$ $x^2 + 12x - 41 = 0$ $(x + 6)^2 - 36 - 41 = 0$ $(x + 6)^2 = 77$ $(x + 6) = \pm\sqrt{77}$ $x = -6 \pm \sqrt{77}$ $x = 2.77$ or -14.77	M1 A1, A1	[3]
24	(a)	$T_n = 2^{n-1} + (3n - 1) + (n + 1)^2$	B1: all 3 terms correct B1: Any correct term	[2]
	(b)(ii)	$T_n - T_{n-1}$ $= 2^{n-1} + (3n - 1) + (n + 1)^2 - [2^{n-2} + (3(n - 1) - 1) + (n)^2]$ $= 2^{n-1} + 3n - 1 + n^2 + 2n + 1 - 2^{n-2} - 3n + 4 - n^2$ $= 2^{n-1} + 2n - 2^{n-2} + 4$ $= 2^{n-2}(2 - 1) + 2n + 4$ $= 2^{n-2} + 2n + 4$	M1: Replace n by $n - 1$ M1: expansion M1	[3]

	(c)	Since the difference of the consecutive terms could be expressed as $2(2^{n-3} + n + 2)$ and n must be a natural number/ positive integer , the difference is always an even number.	B1: must take out factor of 2	[1]
25	(a)	Let r be the radius of the cylinder. $24 = 2\pi r$ $r = \frac{12}{\pi}$ Volume: $\pi \left(\frac{12}{\pi}\right)^2 (17)$ $= \frac{2448}{\pi} \text{ cm}^3$	M1: length = circumference A1	[2]
	(b)	Let the radius of circle be l and radius of the cone be r . Arc length FH : $\frac{1}{4}(2\pi l)$ $\frac{1}{4}(2\pi l) = 2\pi r$ $r = \frac{1}{4}l$ Curved surface area: $\pi \left(\frac{1}{4}l\right) l = 72 \frac{1}{4}$ $l = 17$ Volume of cone: $\frac{1}{3} \pi \left(\frac{1}{4}l\right)^2 \sqrt{289 - \frac{289}{16}}$ $= 311.34438 \text{ cm}^3$ $= 311 \text{ cm}^3$	M1: arc length = circumference M1: Find l M1: height of cone A1	[4]
26	(a)	$\overrightarrow{BD} = \overrightarrow{BA} + \overrightarrow{AD}$ $= 8\mathbf{b} - 4\mathbf{a}$	M1: Find $\overrightarrow{BD}$	

		$\overrightarrow{BE} = \frac{3}{4}(8\mathbf{b} - 4\mathbf{a})$ $= 6\mathbf{b} - 3\mathbf{a}$	A1	[2]
	(b)	$\overrightarrow{BF} = \overrightarrow{BE} + \overrightarrow{EF}$ $= -3\mathbf{a} + 6\mathbf{b} + 2\mathbf{a} - 2\mathbf{b}$ $= 4\mathbf{b} - \mathbf{a}$ $\overrightarrow{BC} = 2(4\mathbf{b} - \mathbf{a})$ $= 8\mathbf{b} - 2\mathbf{a}$	M1: Find $\overrightarrow{BF}$ A1	[2]
	(c)	$\overrightarrow{DC} = \overrightarrow{DB} + \overrightarrow{BC}$ $= 4\mathbf{a} - 8\mathbf{b} + 8\mathbf{b} - 2\mathbf{a}$ $= 2\mathbf{a}$ Since $\overrightarrow{AB} = 2\overrightarrow{DC}$, AB is parallel to DC. ABCD is a trapezium because it has only one pair of parallel lines .	M1: Find $\overrightarrow{DC}$ M1 A1	[3]